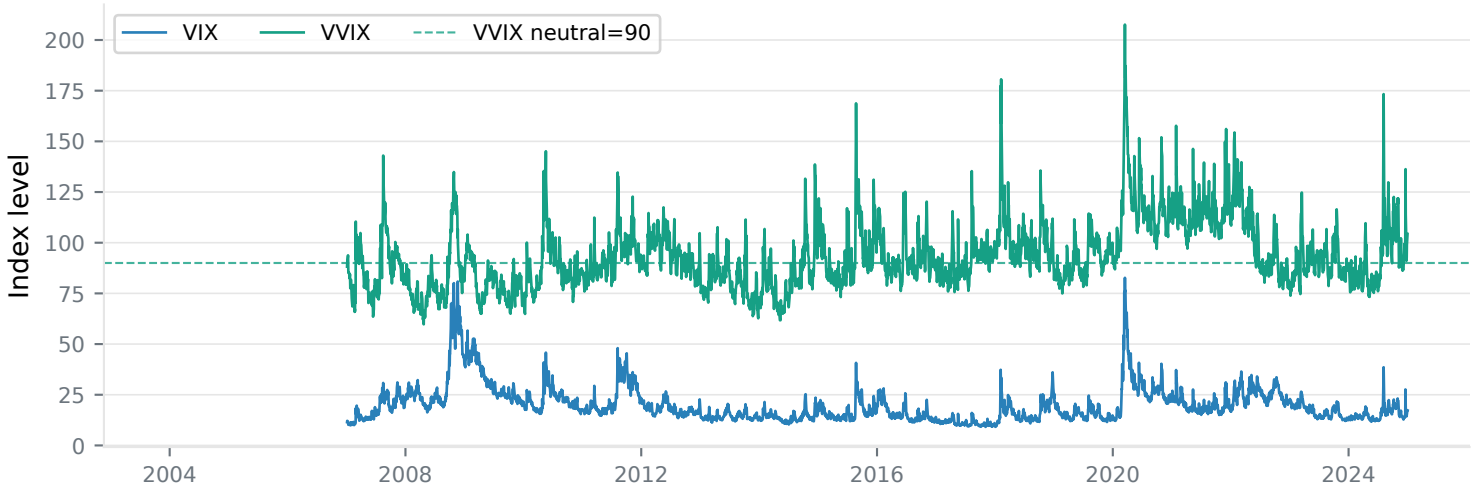
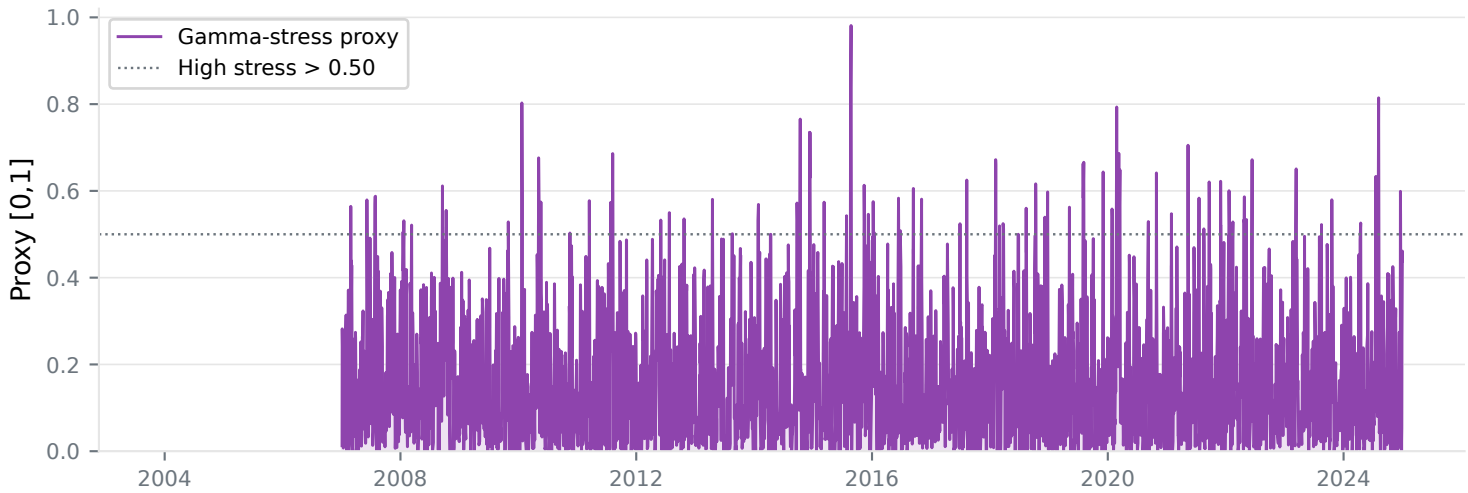
Regime Ensemble - v9.0 Volatility Expansion

SPY 2000-01-01 to 2025-01-01 | New additions: VVIX dampener + gamma-stress proxy

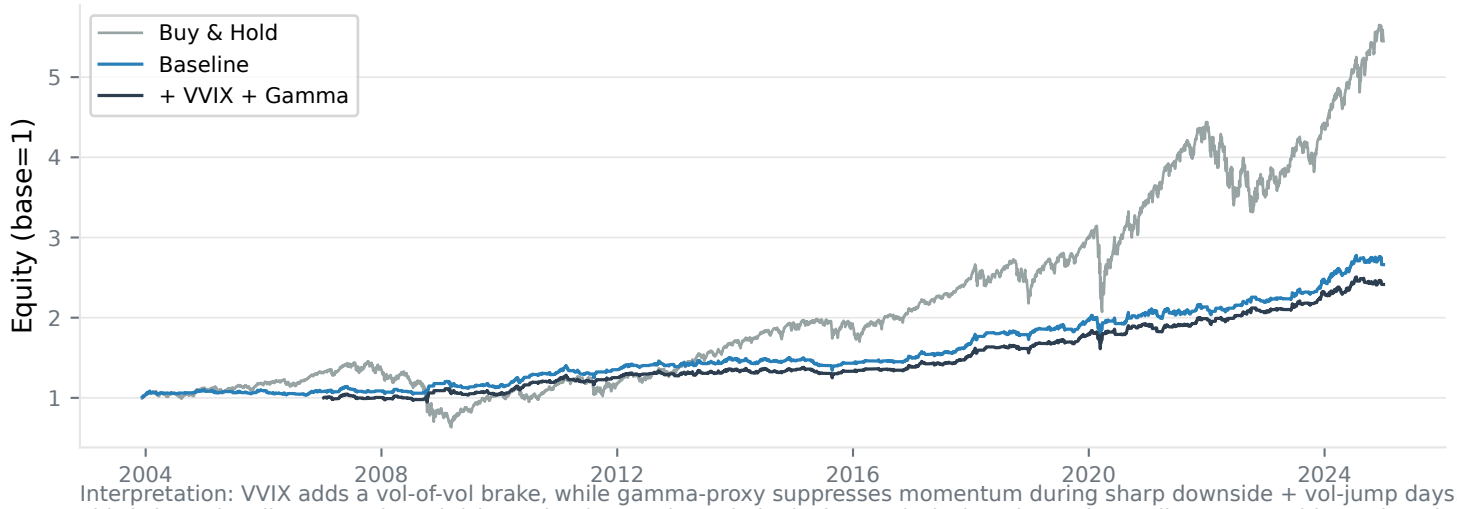
VIX and VVIX Regime Context



Gamma-Stress Proxy (Causal Spot/Vol Shock Composite)



High-Level Equity Impact



Interpretation: VVIX adds a vol-of-vol brake, while gamma-proxy suppresses momentum during sharp downside + vol-jump days.
This is intentionally a proxy-based risk overlay (not options-chain dealer GEX), designed to reduce tail exposure without changing core
VVIX data quality: sufficient for full variant testing.

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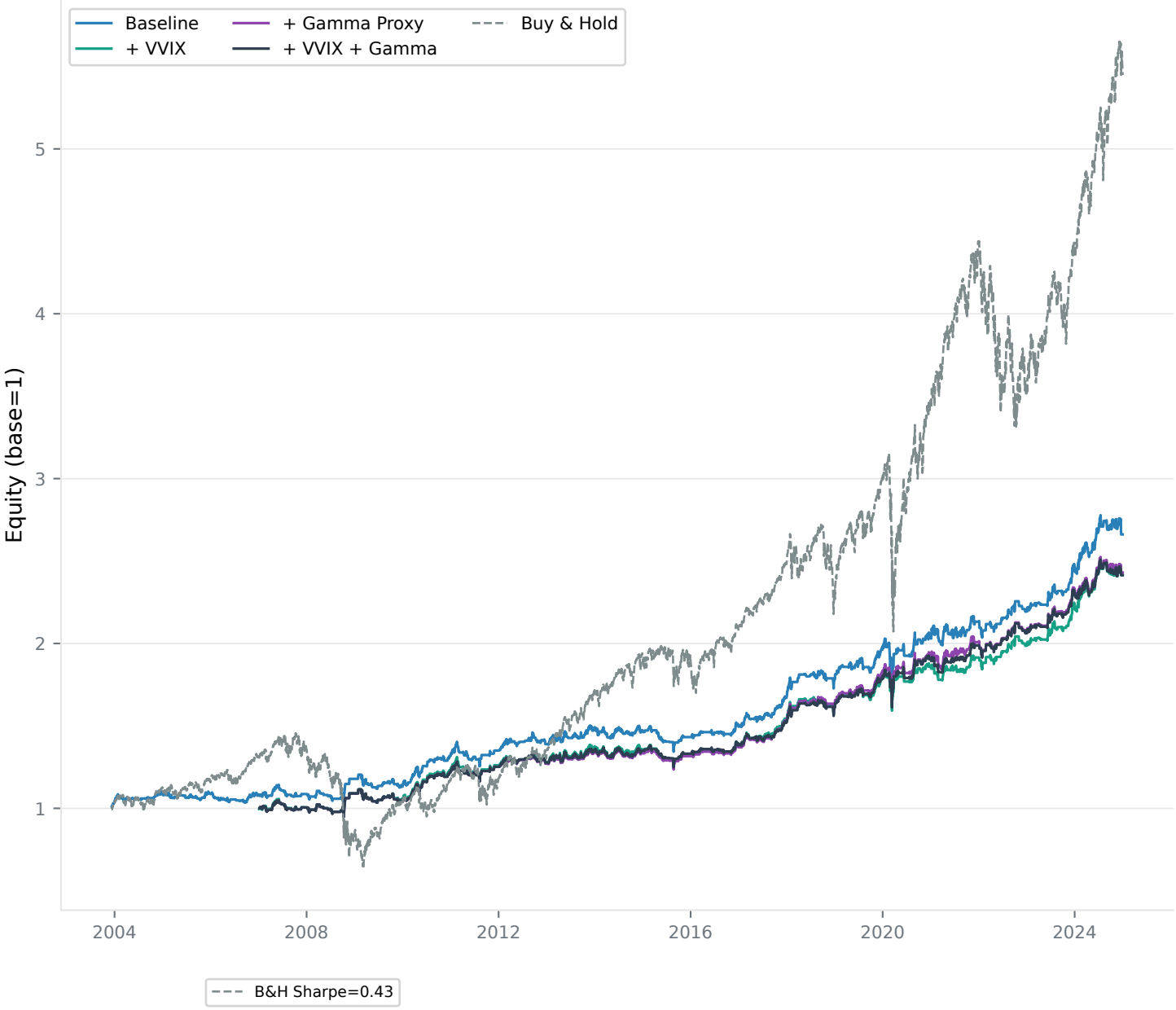
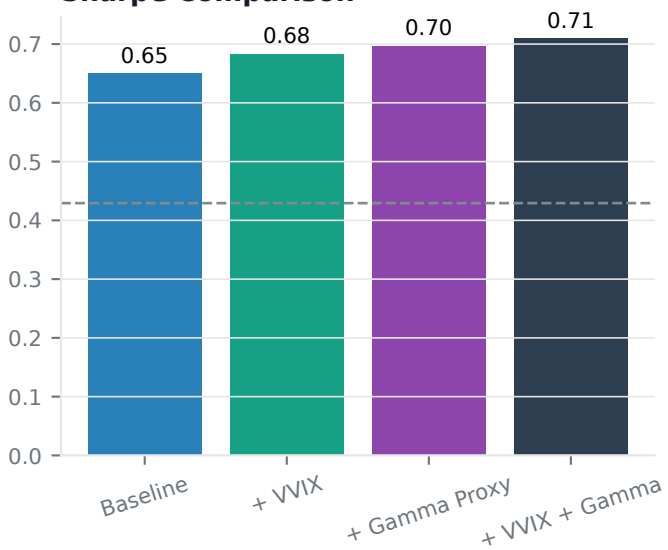
Ablation: Baseline vs VVIX/Gamma Variants (0 bps costs)

Equity Curves



Sharpe Comparison

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Performance Summary (Strategy):

Baseline          Sharpe +0.65  CAGR +4.8%  MaxDD -14.6%
+ VVIX            Sharpe +0.68  CAGR +5.0%  MaxDD -13.6%
+ Gamma Proxy     Sharpe +0.70  CAGR +5.1%  MaxDD -12.5%
+ VVIX + Gamma    Sharpe +0.71  CAGR +5.0%  MaxDD -12.3%

Buy & Hold: Sharpe +0.43, CAGR +8.4%, MaxDD -56.5%

Notes:
- VVIX acts as a steady stress suppressor when
vol-of-vol is elevated.
- Gamma proxy reacts faster to downside spot+vol
shock clustering.
- Combined variant is the strictest risk posture.
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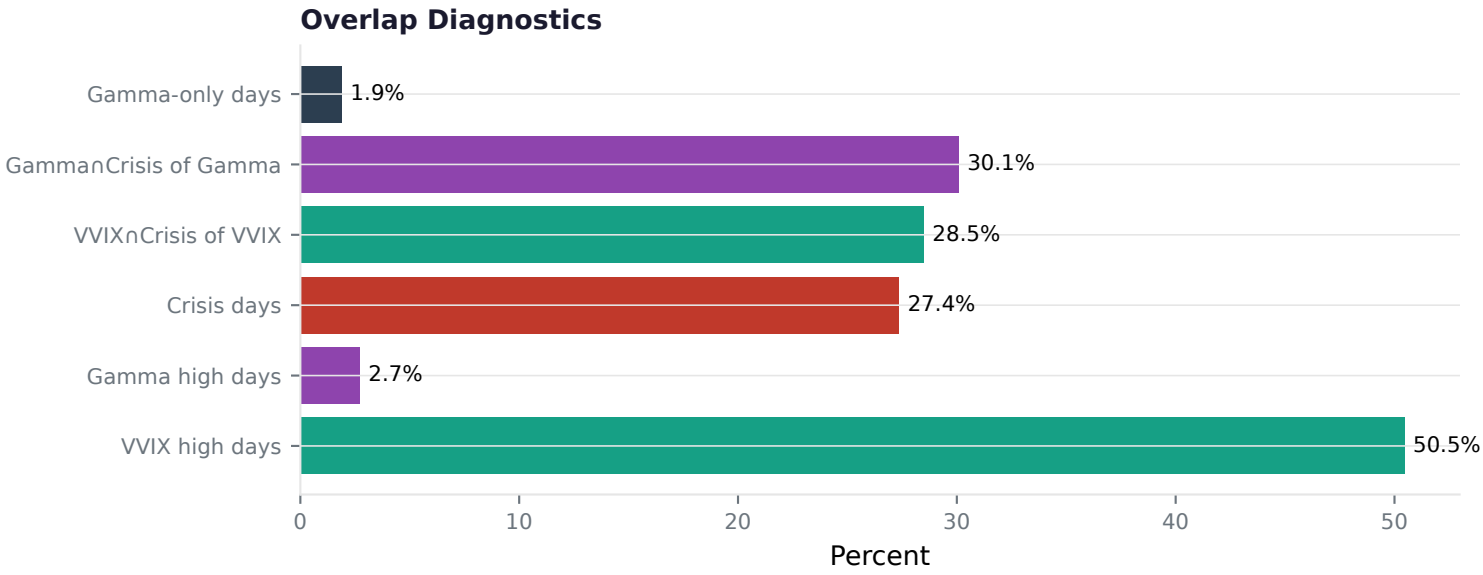
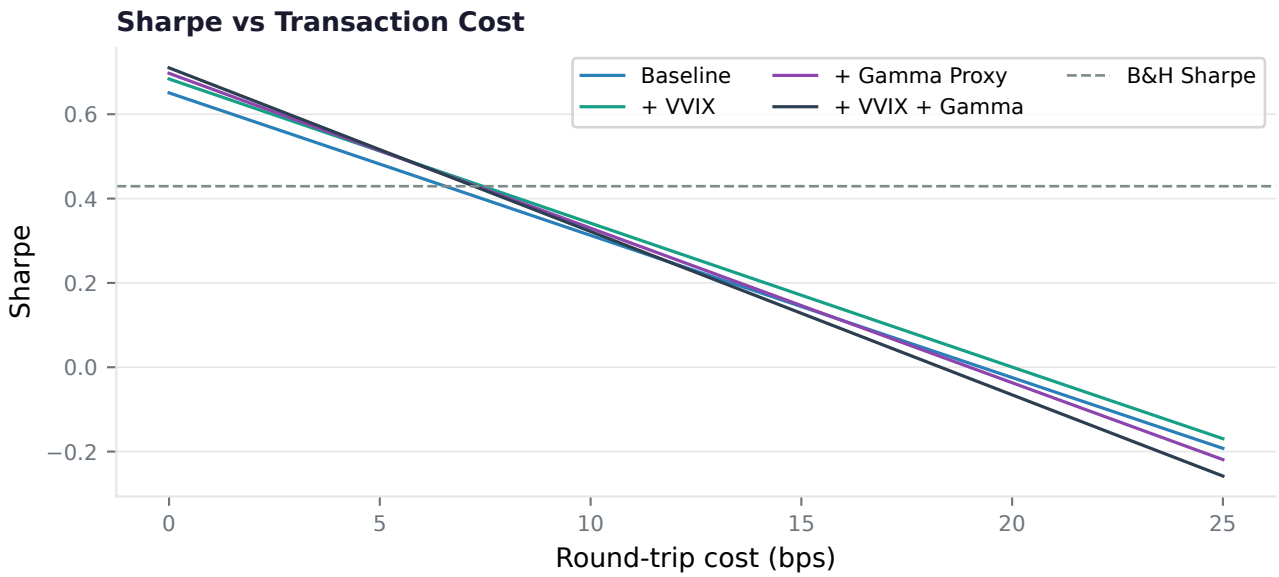
Stress Diagnostics and Transaction-Cost Robustness





Implementation Summary

1. VVIX dampener: multiplies Markov momentum component by a vol-of-vol stress factor.
2. Gamma-stress proxy: causal [0,1] proxy from downside return shock + VIX jump + optional VVIX jump.
3. Both overlays are optional and leave baseline behavior unchanged when disabled.

Observed deltas vs baseline (this sample):
- Sharpe (+VVIX): +0.033
- Sharpe (+Gamma proxy): +0.046
- Sharpe (+VVIX+Gamma): +0.059

Caveat: Gamma proxy is not true dealer GEX from options OI and should be treated as a robust stress overlay, not as exact dealer-position inference.